

The Covariance Decomposition of the Probability Score and Its Use in Evaluating Prognostic Estimates

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The probability score (PS) or Brier score has been used in a large number of studies in which physician judgment performance was assessed. However, the covariance decomposition of the PS has not previously been used to evaluate medical judgment. The authors introduce the technique and demonstrate it by analyzing prognostic estimates of three groups: physicians, their patients, and the patients' decision-making surrogates. The major components of the covariance decomposition—bias, slope, and scatter—are displayed in covariance graphs for each of the three groups. The decomposition reveals that whereas the physicians have the best overall estimation performance, their bias and their scatter are not always superior to those of the other two groups. This is primarily due to two factors. First, the physicians' prognostic estimates are pessimistic. Second, the patients place the large majority of their estimates in the most optimistic category, thereby achieving low scatter. The authors suggest that the calculational simplicity of this decomposition, its informativeness, and the intuitive nature of its components make it a useful tool with which to analyze medical judgment. *Key words:* covariance decomposition; calibration; prognostic estimation; surrogate; probability score. (*Med Decis Making* 1995;15:120–131)

The quantitative analysis and evaluation of physicians' judgment and decision-making performance have been of central concern to researchers and practitioners during the last 30 years. A number of methodologic articles^{1–4} have provided important tools with which to perform such analyses. The goal of this article is to introduce a technique that has not previously been used to evaluate physician judgment but that, we believe, is well suited for that purpose. The technique is the covariance decomposition of the probability score.⁵

The probability score (PS), or Brier score,⁶ has been used in previous studies in which physicians' judgment

performances have been assessed.^{7–13} The PS is calculated in the following way.

Assume that survival is being predicted. Let 0 signify death and 1 signify survival. A physician renders an estimate as to the probability that a person will survive a specified length of time. The difference between this estimate (e.g., 0.75) and the code for the actual outcome (survival, or 1) is calculated (0.25) and squared (0.0625). If a physician makes many such estimates, one for each person in an intensive care unit, for example, then the average PS can be calculated over all such estimates. The resulting PS is a "penalty score," in that larger values of PS signify greater deviations from accurate estimation performance.

Thanks largely to Yates,⁵ physicians have been made aware of a number of ways in which PS can be subdivided into informative components. One common method is the Murphy decomposition¹⁴:

$$\overline{PS} = \bar{d}(1 - \bar{d}) + CI - DI$$

$\bar{d}$ is the base rate of the to-be-predicted event. In our example, this would be the proportion of patients who survived. The other two components of the composition are the calibration index (CI) and the dis-

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crimination index (DI), whose calculations are described in appendix A.

A judge is said to be well calibrated to the extent that his or her judgments correspond to the proportion of times that the predicted event actually takes place. Assume that each of 100 patients is given a 60% one-year survival estimate. If 60% of these patients actually survive one year, then this estimate is perfectly calibrated. Of course, similar levels of accuracy would have to be manifested for all categories of the judge's estimates (e.g., 70%, 80%, etc.) in order for overall calibration to be perfect. The CI merely reflects the extent to which a judge's survival probability assessments within each category deviate from the actual survival rate within that category.

Another component of the Murphy decomposition is the DI. Discrimination refers to the ability to distinguish those occasions when the target event will occur from those occasions when it will not occur. To continue with our survival example, maximal discrimination occurs when all patients who will survive are given a different probability of surviving than all patients who will not survive.

Discrimination is difficult to interpret intuitively, primarily because good discrimination can occur in many different ways. For example, a judge who places all those who survive in the 100% survival estimate category and all those who will not survive in the 0% survival estimate category will have perfect discrimination. However, another judge who mistakenly places all those who survive in the 0% survival estimate category and all those who will not survive in the 100% survival estimate category will also exhibit perfect discrimination. The DI merely reflects the extent to which a judge can distinguish the two groups of patients. The DI is the same whether or not the difference is in the appropriate direction, as the above example illustrates.

Not only is the DI nonintuitive, its graphic depiction is not easily apprehended. As a result, receiver operating characteristic (ROC) curves are a far more common way of graphically depicting discrimination. Thus, one of the Murphy decomposition components, the CI, is often used, whereas the other main component, the DI, is typically replaced with the area under the ROC curve, which correlates highly, but not perfectly, with the DI.¹⁵ The covariance decomposition, on the other hand, provides components that are all easily graphed and apprehended.

The Covariance Decomposition

The covariance decomposition of $\overline{PS}$ is:

$$\overline{PS} = \bar{d}(1 - \bar{d}) + (\text{bias})^2 + [\bar{d}(1 - \bar{d})(\text{slope})](\text{slope} - 2) + \text{scatter}$$

Bias is the difference between the average judgment ($\bar{f}$) and the survival rate ($\bar{d}$). As such, it is a good global measure of judgment accuracy. In the present case, bias would be the difference between the average estimated probability of survival and the actual proportion surviving. Bias may be either positive or negative. Lower absolute values of bias are better; a score of zero indicates no bias. This is a very important statistic, since it can easily be interpreted as a gauge of the optimism or pessimism of the judge.

Since calibration refers to the correspondence between what is predicted and what actually occurs, bias also represents a good, global measure of calibration. It describes the overall difference between the average estimate and the average outcome. Bias is squared in order to obtain the unsigned term referred to as "calibration-in-the-large."¹⁶

Slope is the average probability of survival assigned to those who did survive ($\bar{f}_1$) minus the average probability of survival assigned to those who did not survive ($\bar{f}_0$). Higher slope is better, because this signifies that the judge was able to differentiate between the survivors and the non-survivors.

Whereas bias and slope index the accuracy of the estimates, scatter is an index of the "noisiness" of the estimates. Scatter is a pooled variance derived from the distribution of probability estimates assigned to those who did survive ($\bar{f}_1$) and the distribution of probability estimates assigned to those who did not survive ($\bar{f}_0$). The variance of each distribution is calculated separately before they are pooled. This means that scatter is unrelated to whether the target event occurred. Calculations for scatter, bias, and slope are described in appendix B.

Notice that the covariance decomposition, like the Murphy decomposition, contains the term $[\bar{d}(1 - \bar{d})]$. This is simply the prevalence of the to-be-estimated event (survival, in our example) multiplied by its complement. Since the person performing the estimation task has no control over this term, it is generally not given much attention in the analysis of the judge's estimates.

In a step-by-step fashion we will now construct a covariance graph¹⁶ in order to demonstrate how bias, slope, and scatter can be depicted.

The Covariance Graph

On the abscissa of every covariance graph are the "outcome index" and the base rate ($\bar{d}$). In our example, the two possible states of the outcome index are death (0) and survival (1). Figure 1a, which depicts the bottom "layer" of a covariance graph, contains a vertical dotted line, ($\bar{d}$), that is located at the base rate of survival, which is $\bar{d} = 0.702$. On the ordinate is the "probability judgment." A horizontal dotted line is located at the average probability judgment, $\bar{f} = 0.596$.

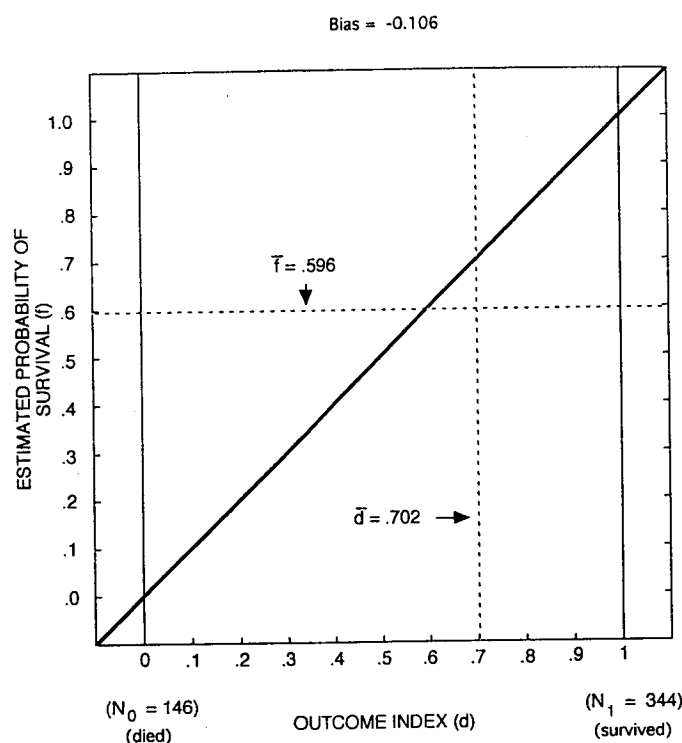


FIGURE 1a. The bottom layer of a covariance graph. $\bar{d}$ signifies survival rate. $\bar{f}$ signifies mean estimate of survival. Points on the dark 45-degree line represent unbiased estimates. Bias corresponds to the vertical distance from the 45-degree line to the point where the $\bar{d}$ and $\bar{f}$ lines cross.

Bias is simply the difference between the average probability estimate and the base rate of the target event. In other words, $\text{bias} = \bar{f} - \bar{d}$. If a judge were unbiased, then the two dotted lines, one horizontal and one vertical, would cross on the dark 45-degree diagonal line in figure 1a. Since the two dotted lines cross below the diagonal in figure 1a, we conclude that this judge is pessimistic. He or she systematically *underestimates* the target event. The bias is -0.106 .

Figure 1b contains the next layer of the covariance graph. On the vertical lines above the 0 and 1 outcome indices are histograms. The bars on these histograms represent the numbers of estimates made at the individual probability categories. In the research described later in this paper, patients were asked to choose among five survival estimates: 90% or better, about 75%, about 50–50, about 25%, or 10% or less. Thus the histogram bars in figure 1b are centered at 0.95, 0.75, 0.50, 0.25, and 0.05, which are the midpoints of the categories. A scale at the top of the covariance graph provides a means of roughly assessing the number of persons represented in each bar of the histogram. To the extent the histograms are spread out along the vertical lines, scatter is large. It is not possible to discern the exact magnitude of scatter by examining the histograms, but it is often possible to make rough comparisons between two graphs in order to determine

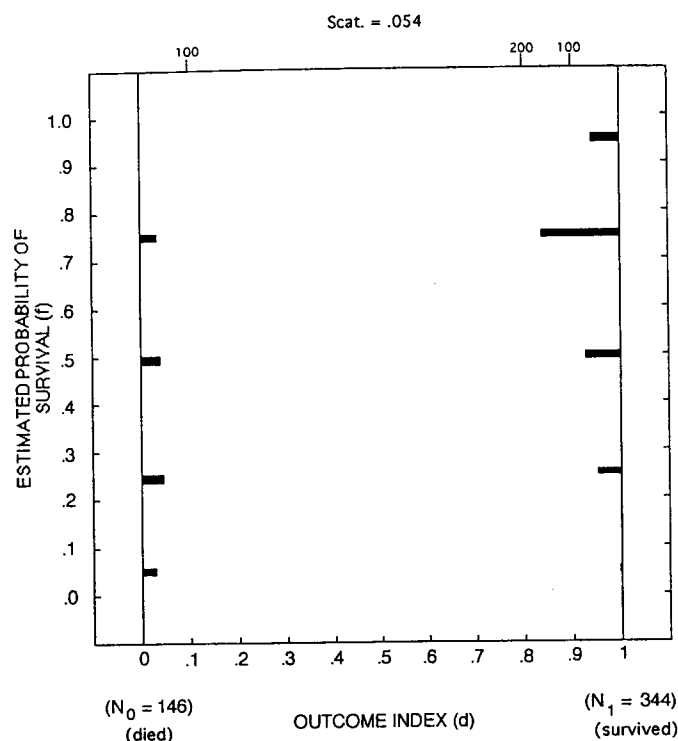


FIGURE 1b. The second layer of a covariance graph. The histograms represent the numbers of estimates at individual probability categories. The scale at the top of the graph provides a means of gauging the number of estimates in each category. This provides a rough measure of scatter.

which has the greater scatter.

Figure 1c depicts the final layer of the covariance graph. On the outcome index line extending vertically from 1 is a dot with the label " $\bar{f}_1$." This is the average estimate given to those patients who survived. On the outcome index line extending vertically from 0 is a dot with the label " $\bar{f}_0$." This is the average estimate given to those patients who died. The slope of the regression line going through these two points is equal to $\bar{f}_1 - \bar{f}_0$. If a judge gave approximately the same average estimate to those who survived and those who did not, the regression line would be flat. The extent to which the regression line diverges from the horizontal represents the degree to which the estimator was able to discriminate the two groups. Thus slope is an easy way of gauging discrimination. In our example, the slope is $(0.677) - (0.408) = 0.269$.

We combine the layers in figure 1d, which contains a complete covariance graph. At the top of our covariance graph we have placed the magnitude of each of the components of the decomposition.

In summary, the covariance graph enables one to view a global measure of calibration (bias), a measure of discrimination (slope), and a depiction of variability (scatter), as well as statistics that describe the average outcome ($\bar{d}$), the average estimate irrespective of outcome ($\bar{f}$), and the average estimates given to the two

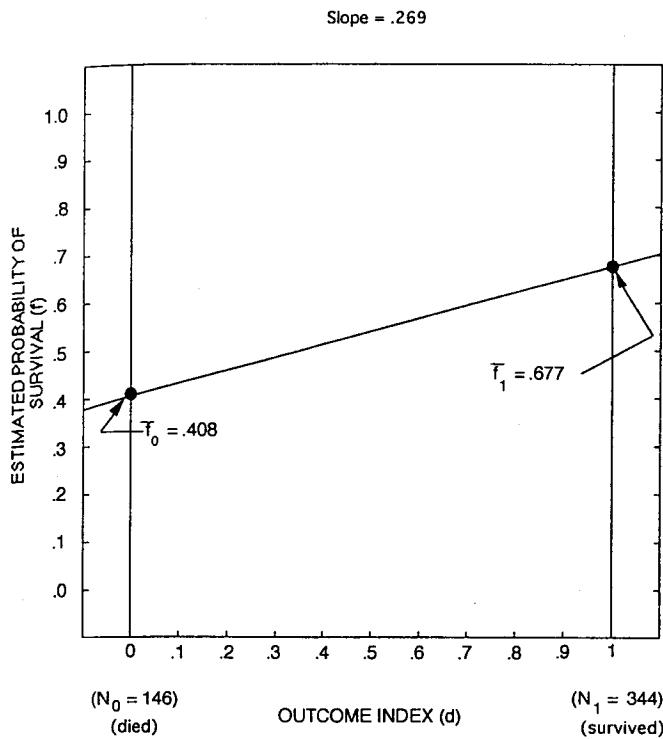


FIGURE 1c. The third layer of a covariance graph. $\bar{f}_0$ represents the average estimate given to those patients who died. $\bar{f}_1$ represents the average estimate given to those patients who survived. The regression line connecting these two points depicts the slope.

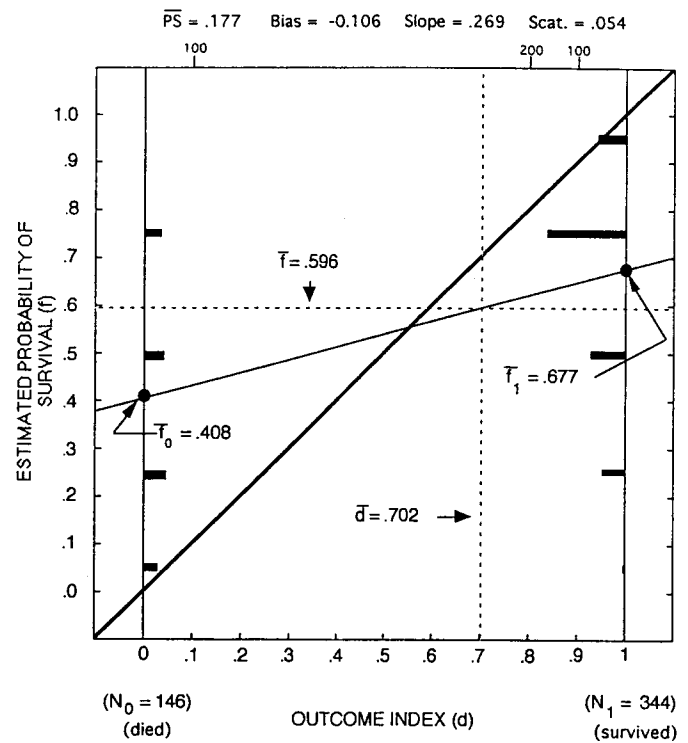


FIGURE 1d. The complete covariance graph.

outcome groups ($\bar{f}_1$ and $\bar{f}_0$). None of these statistics requires much calculation, and all are easily constructed on regular graph paper.

We will now demonstrate the use of the covariance decomposition on a complete set of data.

Method

PATIENT POPULATION

The patients eligible to participate in this research were part of a larger, multisite, research program involving 4,301 patients. Detailed methods used for this larger study (SUPPORT) have been reported elsewhere.¹⁷ All seriously ill patients presenting at any of five hospitals between June 12, 1988, and June 12, 1991, and meeting the criteria¹⁷ for one of the following nine diagnoses were eligible for inclusion in SUPPORT: acute respiratory failure, chronic obstructive pulmonary disease, congestive heart failure, cirrhosis, coma, colorectal cancer metastatic to liver, non-small-cell lung cancer stage III or IV, multiple-organ-system failure and malignancy, or multiple-organ-system failure and sepsis. Patients were ineligible to enter SUPPORT if they met the exclusion criteria documented elsewhere.¹⁷

In order for a patient to be included in the current study, it was necessary to obtain prognostic estimates from that patient, his or her physician, and the designated surrogate. If any of these persons did not or could not provide an estimate of survival, the patient and the corresponding physician and surrogate were excluded from the study reported here.

PROCEDURE

On approximately the third day after the patient's entry into SUPPORT (interview window: day 2-6), an interviewer entered the patient's room and requested the patient's consent to conduct an interview. If the patient agreed, the interview commenced. Among the questions asked were the following:

"What are the chances that you will live for two months or more if the current plan of care stays the same?"

"Are the chances 90% or better, about 75%, about 50-50, about 25%, or 10% or less that you will live for two months or more? Please give me your best estimate."

"How about six months or more?"

The interviewer also asked the patient:

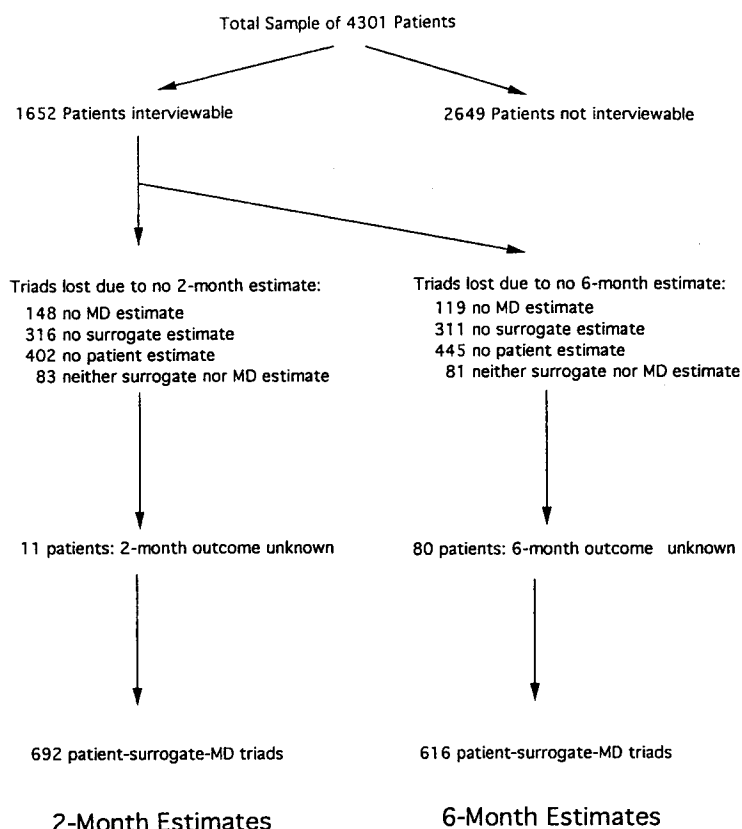


FIGURE 2. Reasons for patient attrition.

"Who would you want to help your physician make a decision about your medical care if you were ever to ill to do so? (If you had to choose, which one person would you want to make these decisions on your behalf?)"

The patient was asked for permission to speak to this person. If the patient gave his or her consent, an attempt was made to conduct an interview with this decision-making surrogate on that same day. The surrogate was first asked whether he or she was the person who would make decisions about the care of the patient if the patient were not able to do so. Only after the surrogate had verified his or her status did the interviewer continue. Among the questions asked were the same ones asked of the patient concerning the probability of the patient's living two and six months.

On approximately the third day after the patient's entry into SUPPORT (interview window: day 2–6), the patient's physician was also interviewed. Among the questions asked were the following: "What is the probability that the patient will live for two months or more?" "What is the probability that the patient will live for six months or more?" The physician gave a number from 0 to 100 to answer each question. To make them comparable to the patient and surrogate estimates, the physicians' responses were grouped such that they corresponded to the five patient and sur-

rogate response categories: estimates greater than or equal to 90% were grouped into one category, 65% to 89% comprised the next category, 36% to 64% the next, 11% to 35% the next, and estimates less than or equal to 10% comprised the last category. For all three groups, the analyses required the use of points rather than ranges for each category. Thus all estimates within each category were converted to the approximate midpoint for each category: 95%, 75%, 50%, 25%, and 5%.

Results

PATIENT CHARACTERISTICS

The total number of patients enrolled in SUPPORT—the research program from which these patients were drawn—was 4,301. Since patients could participate in this particular study only if they, their physicians, and their surrogates all could provide the necessary prognostic estimates, the most critically ill and debilitated patients were less likely to be included, since such patients were often unconscious or on mechanical ventilation and could not be interviewed. Indeed, the 692 patients whose data were used for the two-month estimates were younger (mean age 60.8 years) and were less severely ill (mean APACHE III¹⁸ score 29.1) than the 3,609 patients in SUPPORT whose data were not used for the two-month estimates (mean age 62.8 years and mean APACHE III score 38.3). In figure 2 we depict the reasons for the exclusion of various categories of patients and how we arrived at our final two-month estimation sample of 692 patient-surrogate-physician "triads." As can be discerned from figure 2, approximately three-fourths of the excluded patients could not participate in this study because they could not be interviewed.

The two-month survival rate for the 692 patients who gave two-month estimates was 84.7%. Compared with 949 SUPPORT patients not included in the current report but who were interviewed and (whose survival status was determined, the 692 patients whose data were included did not differ significantly from excluded patients on APACHE III score, number of comorbidities, and number of days in the hospital before they entered the study.*

Of those 692 patients who gave two-month prognostic estimates, not all gave usable data for the six-month estimate. Figure 2 depicts the manner in which we obtained the total number of 616 six-month-estimate patient-surrogate-physician triads. The six-month survival rate of these 616 patients was 69.8%.

Compared with patients who were not included in the study but who were interviewed and whose survival status was determined, the 616 patients whose six-month data were included in this part of the study

did not differ significantly from excluded patients on age, APACHE III score, number of comorbidities, average treatment intensity score, number of days in the hospital before they entered the study, or six-month survival rate.

Table 1 displays the disease categories of the 692 patients who gave two-month estimates and the 616 patients who gave six-month estimates.

We were almost entirely successful (>99.8%) in interviewing the surrogates on the same day as the patients.

QUALITY OF ESTIMATES

Table 2 contains the probability score, bias, slope, and scatter measures for all two-month and six-month judges. Figures 3a, b, and c depict these measures in covariance graph form.

In general, the physicians were superior to the other two groups in their overall PS. However, the physicians were not superior to the other groups on every component of the PS. One interesting example is bias. Consider first the two-month estimates. Reference to figure 3a reveals that the patients' average two-month estimate ($\bar{f}$) and the two-month base-rate of survival ($\bar{d}$) crossed almost exactly on the 45-degree line. This means that the patients' estimates had negligible bias, which is confirmed by the data presented in table 2. Figure 3b shows that surrogates were somewhat pessimistic. This corresponds to the bias of -0.037 listed in table

*The 692 included patients did differ significantly from certain subsets of the excluded patients. The 692 included patients had a significantly ($p < 0.05$) higher six-month survival rate (70.2%) than the 402 patients who were interviewed but did not give a two-month estimate (59.0%). Included patients also had a significantly higher treatment intensity score (TISS)¹⁹ (17.1) than the 83 patients whose physicians and surrogates both failed to make prognostic estimates (14.6), and they were significantly younger (60.8 years) than two groups: the 402 patients who were interviewed but did not give a two-month estimate (65.0 years) and the 148 patients whose physicians made no six-month estimate (62.2 years). Also, it should be noted that this report was written before the completion of the SUPPORT study. Therefore the small proportion of patients whose final outcomes were unknown at the time this report was written will eventually be included in one of the outcome categories.

Table 1 • Disease Categories of Patients Who Gave Two-month Estimates and Patients Who Gave Six-month Estimates

Disease Category	2-month Estimators	6-month Estimators
Acute respiratory failure	116	106
Chronic obstructive pulmonary disease	83	76
Congestive heart failure	199	183
Cirrhosis	79	74
Coma	1	1
Colon cancer	66	54
Lung cancer	86	69
Multiple-organ-system failure + malignancy	23	19
Multiple-organ-system failure + sepsis	39	34
TOTAL	692	616

2. Figure 3c shows that the physicians were more pessimistic, as evidenced by the fact that the $\bar{f}$ and $\bar{d}$ lines cross substantially beneath the 45-degree line. In fact, the physicians were approximately 12% too pessimistic in their two-month estimates.

Notice that whereas the physicians were too pessimistic when their estimates are viewed in the aggregate, their slope indicates that they were far more discriminating than either the patients or the surrogates. This is easily seen by comparing the slopes of the lines connecting $\bar{f}_1$ and $\bar{f}_0$ in figures 3a, b, and c. The patients have a much flatter slope than either the surrogates or the physicians, which is again confirmed in the data presented in table 2.

Another index of estimation quality on which the physicians were not clearly superior to the other two groups was "scatter," which is a measure of estimation variability unrelated to patients' actual survival. A comparison of the patterns of the histograms in figures 3a, b, and c suggests that the surrogates had more scatter in their estimates than did the other two groups, which in turn were similar to each other. Consulting Table 2 confirms that the physicians' two-month estimates were slightly more variable than those of the patients

Table 2 • Mean Probability Score ($\overline{PS}$), Mean Estimate ($\bar{f}$), Mean Estimate Assigned to Survivors ($\bar{f}_1$), Mean Estimate Assigned to Non-Survivors ($\bar{f}_0$), Bias, Slope, and Scatter for All Two-month and Six-month Estimators*

	$\overline{PS}$	$\bar{f}$	$\bar{f}_0$	$\bar{f}_1$	Bias	Slope	Scatter
Two-month estimates ($n = 692$)							
Patients	0.146	0.864	0.790	0.877	0.017	0.087	0.038
Surrogates	0.138	0.810	0.648	0.839	-0.037	0.191	0.052
Physicians	0.134	0.726	0.549	0.758	-0.121	0.209	0.039
Six-month estimates ($n = 616$)							
Patients	0.234	0.832	0.745	0.870	0.134	0.125	0.054
Surrogates	0.206	0.758	0.604	0.824	0.060	0.220	0.075
Physicians	0.180	0.592	0.412	0.670	-0.106	0.258	0.053

*Base rate $\bar{d} = 0.847$ for two-month survival. Base rate $\bar{d} = 0.698$ for six-month survival.

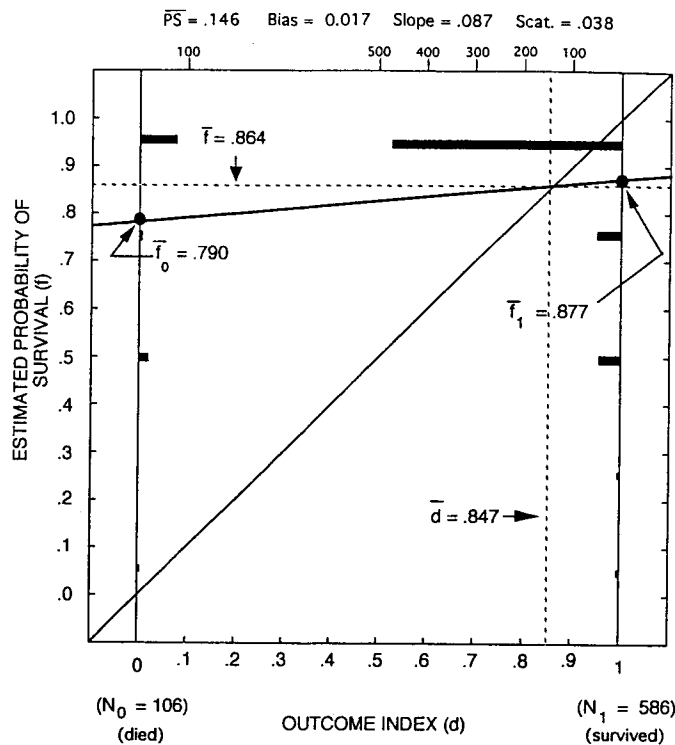


FIGURE 3a. Covariance graph for patients who made two-month estimates.

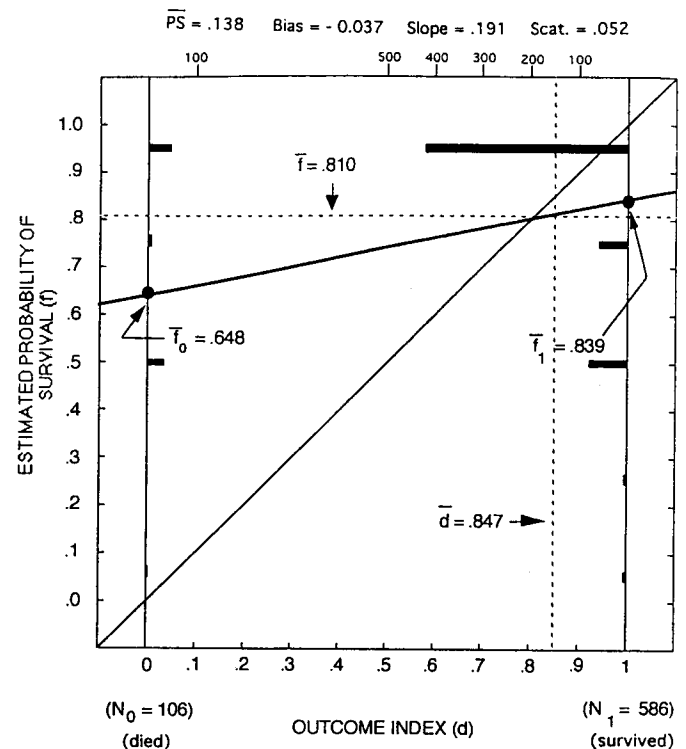


FIGURE 3b. Covariance graph for surrogates who made two-month estimates.

but less variable than those of the surrogates.

The histograms along the left and right margins of the panels in figure 3 depict the actual numbers of estimates made in the individual categories. These graphs indicate that the patients and the surrogates placed a large majority of their estimates in the most optimistic category. Such behavior is particularly detrimental to slope. The patients' data show inferior slope, because the patients who survived and those who did not *both* tended to place the bulk of their estimates in the highest category ($\geq 90\%$). Thus, the mean estimate of survival made by those who did survive exceeded the estimate of those who did not survive by only 0.087 ($0.877 - 0.790$).

The histograms in figure 3c, which depict the physicians' data, show that they used the less optimistic categories for the non-surviving patients more than those patients themselves did (fig. 3a), resulting in greater slope.

The histograms also provide valuable information with regard to the source of the physicians' negative bias. Since the physicians used the less optimistic categories more than did the other two groups and were frequently incorrect for doing so, the physicians' bias was much more negative than the biases of the other two groups.

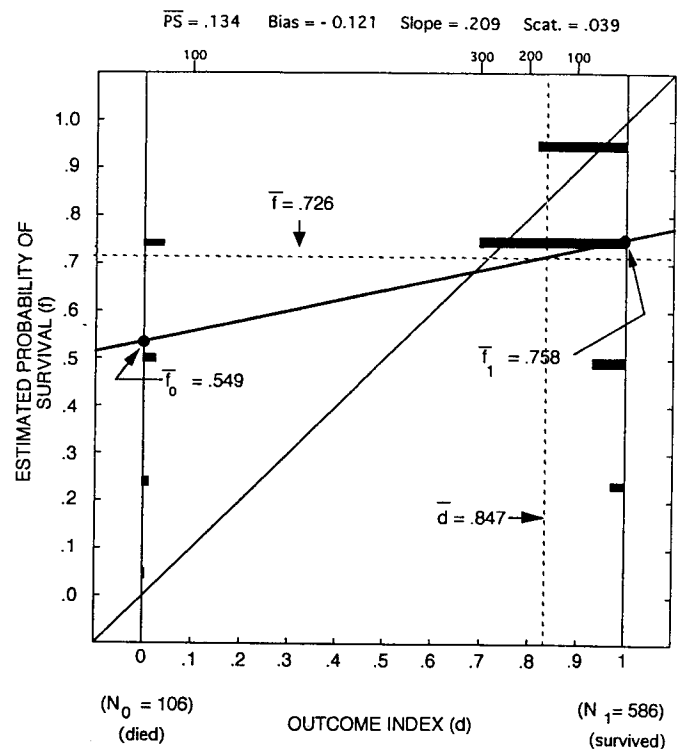


FIGURE 3c. Covariance graph for physicians who made two-month estimates.

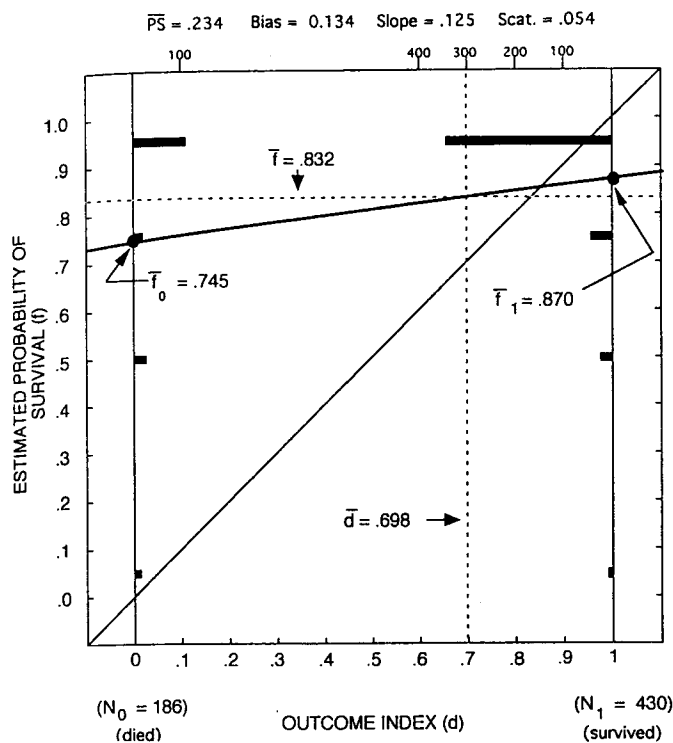


FIGURE 4a. Covariance graph for patients who made six-month estimates.

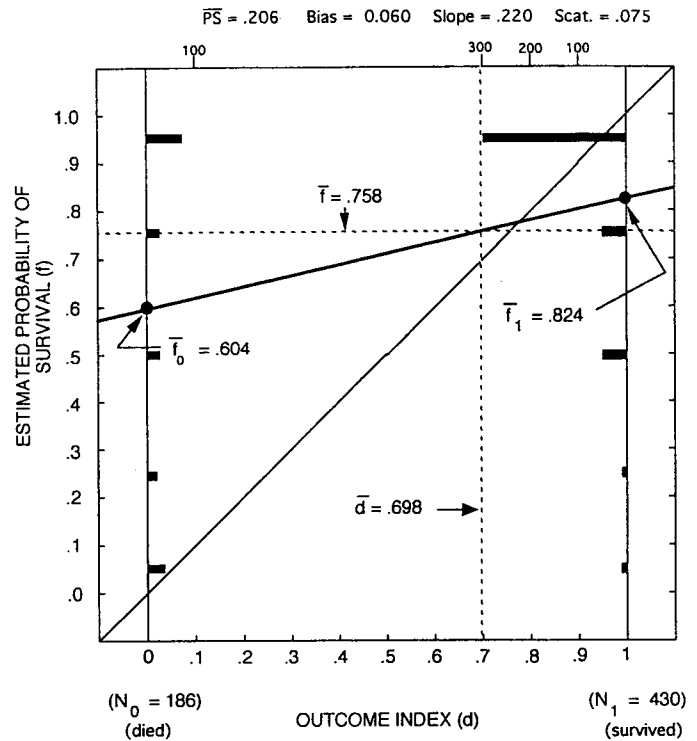


FIGURE 4b. Covariance graph for surrogates who made six-month estimates.

Covariance graphs for six-month survival are shown in figures 4a, b, and c, respectively. These figures resemble figures 3a, b, and c. However, the patients' graph does show more positive bias in their six-month estimates compared with their two-month ones; the $\bar{d}$ and $\bar{f}$ lines cross well above the 45-degree line. The physicians' six-month estimates were again pessimistic, as evidenced by the $\bar{f}$ and $\bar{d}$ lines crossing substantially below the 45-degree line. The surrogates were the least biased among the six-month estimators.

All groups appear to have had greater scatter in their six-month estimates than in their two-month ones. This is understandable, since all of the groups were attempting to predict an outcome that was further in the future.

Since the group of 692 patients who gave two-month estimates was obviously not identical to the group of 616 who provided six-month estimates, comparisons between the two-month and six-month estimates are slightly tainted by subject differences. Therefore, we calculated all the indices of judgment quality among the 585 patients, their physicians, and their surrogates who gave both two-month and six-month estimates. These data are extremely similar to those depicted in table 2; therefore, we have not presented them in a separate table.

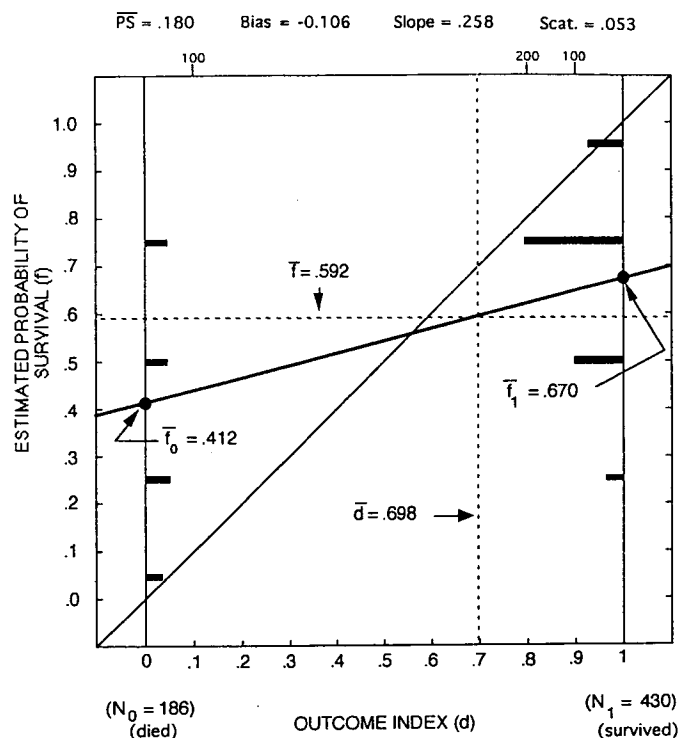


FIGURE 4c. Covariance graph for physicians who made six-month estimates.

Discussion

The covariance decomposition and the accompanying graphs help provide a context in which the estimation performances of the three groups can be understood. For example, the physicians were not superior to the other two groups on the indices of bias and scatter. How is this possible?

The histograms on the covariance graphs show that the patients placed approximately three-fourths of both their two-month and six-month estimates in the category "90% or better." This skewed distribution of patients' estimates is evident in figures 3a and 4a, in which the histogram bars corresponding to the most optimistic estimate are much larger than the other bars. The overuse of the most optimistic category ($\geq 90\%$) had a number of consequences for the data in table 2.

First, the patients' scatter was diminished by the fact that so many of their estimates were in a single category. In fact, this resulted in smaller scatter in the estimates of the patients compared with that of the physicians in the two-month data.

Second, the preponderance of optimistic estimates directly influenced bias in at least two ways. Patients' bias was low in the two-month estimates, since the survival rate was 84%, which is relatively close to the most optimistic category. The strong tendency of patients to place their estimates in that category thus helped to minimize two-month bias. However, this same factor tended to increase bias in the patients' six-month estimates. This is due to the fact that the six-month survival rate was only 70%. This was substantially beneath the most optimistic category, which was also the modal response for patients making six-month estimates.

The third consequence of the overuse of optimistic estimates by the patients was diminished slope. Both survivors and non-survivors gave generally high estimates, even though the non-survivors' more serious conditions did not warrant such optimism. The shallow slope of the patients' estimates compared with the slope of the other two groups' estimates is particularly apparent.

The surrogates used the most optimistic category less often than the patients did—only 68% of the time for the two-month estimates and 61% for the six-month data. Their positive bias was therefore not as large as that of the patients, and their broader use of more categories improved their slope.

The physicians' data departed substantially from those of the other two groups. The physicians used the most optimistic category only 27% of the time in their two-month estimates and only 13% of the time in the six-month data. This resulted in negative bias, since approximately 85% of the patients actually survived two months, and 70% survived six months. The

fact that the physicians did not overuse the most optimistic category was related to their ability to discriminate survivors from non-survivors to a far greater extent than the other two groups were able to do. The physicians' discrimination ability is reflected in their superior slope.

It should be noted that all of these conclusions are easily apprehended from the covariance graph, which displays the items generated by the covariance decomposition. A major difference between the Murphy and covariance decompositions is that the latter provides components that lend themselves to between-group comparisons. For example, if physicians have a steeper slope than patients do, this implies that the physicians were more likely than the patients to have given higher survival estimates to survivors than to non-survivors. On the other hand, if the physicians had a larger DI than the patients, we would not know whether the physicians were more likely than the patients to have given higher survival estimates to survivors than to non-survivors. The DI is relatively opaque, unlike slope, which is intuitive.

BASES FOR DIFFERENCES IN BIAS, SLOPE, AND SCATTER

Bias. Yates¹⁶ enumerated three influences on the magnitude of bias: incentives, biased information sources, and lack of information about the base rate. All three are possible in this data set.

It has been suggested that physicians tend to "hang crepe"²⁰ by giving pessimistic survival estimates. An incentive for such pessimism is that family members will not object if an outcome is more positive than anticipated but may be very upset if an outcome is more negative. One way for a physician to guard against family anger is to prepare them for a poor outcome. Low survival estimates may be a manifestation of this strategy and would therefore be a possible cause of physicians' negative bias.

One piece of evidence congruent with this conjecture is the fact that the same pessimism was evident in the physicians' two- and six-month estimates: the physicians were sensitive to the fact that the six-month survival rate was approximately 15% lower than the two-month one. This is consistent with the presence of a stable incentive to render pessimistic estimates. The surrogates and particularly the patients seriously underappreciated the fact that the six-month survival rate was much lower than the two-month one. The patients lowered their estimates only about 3%.

A biased information source may have been an influence on the surrogates' data. Notice that the surrogates' bias is closer to that of the patients than that of the physicians. This may be due to the fact that the surrogates probably had more contact with the patients than with the physicians. The patient's opti-

mism may thus have influenced the surrogate more than the physician's pessimism. An alternative explanation is that the patient and the surrogate may have had similar psychological incentives for optimism.

Since the patients and the surrogates could not be expected to know the base rates of survival for the various disease categories, the physicians should have enjoyed a substantial advantage in knowledge. Indeed, Christensen-Szalanski and colleagues²¹ have shown that physicians are superior to laypersons in estimating the mortality rates of various diseases. However, the physicians' median error in that study was 150%. Therefore, while physicians' knowledge of base rates may be much better than that of laypersons, their knowledge may still be imperfect. Since both physicians and laypersons overestimated the mortality rates in that study, it is unlikely that lack of knowledge of base rates could by itself account for the pattern of bias found in the present study, since in our data the patients were optimistic, whereas the physicians were pessimistic.

Slope. Yates¹⁶ suggests that discrimination skill may be the most important determinant of slope. In order to assign higher survival estimates to those who actually survive, one must know what cues predict survival. Again, physicians should have substantially more knowledge than the other two groups, which should give them an enormous advantage. Physicians have previously viewed patients who survived and those who did not. Presumably physicians would have learned from such feedback and would have discovered which cues were associated with each outcome. Patients would have had drastically fewer such learning opportunities.

A second reason physicians would be expected to have higher slopes than patients is that the physician knows a great deal more diagnostic information about each patient than the patient knows. For example, the physician knows how to interpret laboratory results and radiographic information. The patient knows only the subset of these data the physician is willing and able to communicate. To the extent the physician is able to use this informational advantage, he or she should also have a better slope than the patient.

Denial^{22,23} might be a third factor responsible for the patients' shallow slope. The awareness of diagnostic internal cues may be blunted by a patient who is trying to deny their existence or importance. By rendering such cues less salient, a patient may thereby be lessening the impact of a cue that would otherwise have contributed to improved slope. Indirect evidence consistent with the role of denial is that the surrogates' slope is closer to that of the physicians than that of the patients. Presumably the surrogates did not have greater knowledge than the patients with which to achieve their better discrimination performance. However, one factor that may have placed the surrogates

in an advantageous position was that the surrogates, in general, were not seriously ill. Therefore they would have had less egocentric involvement in the illness, and consequently they might have had lower motivation for denial. This might have resulted in greater awareness of the importance of the available cues.

Scatter. Yates¹⁶ lists reliance on weak cues and inconsistency as potential contributors to scatter.

Other researchers²⁴⁻²⁶ have shown that experts may use weak or nondiagnostic cues when making judgments. For example, compared with novices, experienced stockbrokers attempt to use more cues.²⁴ Some of these cues are not related to the to-be-predicted event. As a result, reliance on such cues introduces "noise" into the system, thereby increasing scatter.

Other researchers²⁷⁻²⁹ have shown that physicians do not use cues in a perfectly consistent way. Of course, patients and surrogates might be expected not to do so either. Therefore, it is unlikely that this factor alone is responsible for the patients' surprisingly low scatter. Undoubtedly their marked propensity to endorse the most optimistic category was a major cause.

Conclusion

We suggest that the covariance decomposition and the covariance graph provide a highly satisfactory means of assessing physician judgment. Both can be done quickly, and they are highly informative. The steps in constructing the graph are simple:

1. The mean outcome ($\bar{d}$) is depicted with a vertical line intersecting the abscissa at the appropriate location.
2. The mean estimate ($\bar{f}$) is depicted with a horizontal line intersecting the ordinate at the appropriate location.
3. Bias is either positive, negative, or zero, depending on whether the $\bar{f}$ and $\bar{d}$ lines intersect above, below, or on the 45-degree line. Bias², which is $(\bar{f} - \bar{d})^2$, is calibration-in-the-large.
4. The average estimate given to one outcome ($\bar{f}_0$) and the average estimate given to the other outcome ($\bar{f}_1$) are placed on the 0 and 1.0 lines, respectively.
5. The slope of the regression line connecting $\bar{f}_0$ and $\bar{f}_1$ represents slope, which is a measure of discrimination.
6. The histograms on the left and right margins depict scatter.

Once the covariance decomposition is calculated, individual or group comparisons on these important measures can readily be made, and the depiction of such differences in the covariance graphs can easily be done.

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APPENDIX A

Calculation of the Murphy Decomposition

The Murphy decomposition* of the average probability score (PS) is:

$$\overline{PS} = \bar{d}(1 - \bar{d}) + CI - DI.$$

where $\bar{d}$ is the base rate of the to-be-predicted event.

The calibration index (CI) is calculated in the following way. Assume that the survival estimates are grouped into the categories we used. Estimates greater than or equal to 90% were grouped into one category, 65% to 89% comprised the next category, 36% to 64% the next, 11% to 35% the next, and estimates less than or equal to 10% comprised the last

category. All estimates within each category were converted to the approximate midpoint for that category: 95%, 75%, 50%, 25%, and 5%. Within each category we calculated the survival rate of patients assigned by the judge to that category. For example, 456 patients estimated that they would have a 90% or greater chance of surviving six months. Their survival rate was not 95% (the mid-point of the "90% or greater" category) but instead was 75.2%. This difference (0.950 - 0.752) is squared and multiplied by the number of patients in this category. This is done for each of the j judgment categories. The resulting calibration indices for the five categories are added together and divided by the total number of patients. This is the overall calibration index (CI). Thus,

$$CI = \sum CI_j/N$$

where the CI for each of the j categories is

$$CI_j = N_j(f_j - \bar{d}_j)^2$$

where N_j is the number of patients in category j , f_j is the midpoint of that category, and $\bar{d}_j$ is the survival rate of the patients assigned to that category.

The discrimination index (DI) is calculated in the following way. Within each of the j categories square the difference between the overall survival rate and the survival rate within that category. Multiply this number by the number of patients within that category. This is the discrimination index for that category. Sum these indices for the five categories and divide by the total number of patients to obtain DI. Thus,

$$DI = \sum DI_j/N$$

where the DI for each of the j categories is

$$DI_j = N_j(\bar{d}_j - \bar{d})^2$$

APPENDIX B

Calculation of the Covariance Decomposition

The covariance decomposition of $\overline{PS}$ is:

$$\overline{PS} = \bar{d}(1 - \bar{d}) + (\text{bias})^2$$

$$+ [\bar{d}(1 - \bar{d})(\text{slope})](\text{slope} - 2) + \text{scatter}$$

where $\bar{d}$ is the base-rate of the to-be-predicted event.

Bias is the difference between the average judgment ($\bar{f}$) and the survival rate ($\bar{d}$).

Slope is the average probability of survival assigned to those who did survive ($\bar{f}_1$) minus the average probability of survival assigned to those who did not survive ($\bar{f}_0$).

Scatter is defined as:

$$\frac{N_1 \text{Var}(f_1) + N_0 \text{Var}(f_0)}{N_1 + N_0}$$

As the formula indicates, the variance of the estimates assigned to those who did survive is weighted by the number of patients who survived (N_1). The variance of the estimates assigned to those who did not survive is weighted by the number of patients who did not survive (N_0).

did not differ significantly from excluded patients on age, APACHE III score, number of comorbidities, average treatment intensity score, number of days in the hospital before they entered the study, or six-month survival rate.

Table 1 displays the disease categories of the 692 patients who gave two-month estimates and the 616 patients who gave six-month estimates.

We were almost entirely successful (>99.8%) in interviewing the surrogates on the same day as the patients.

QUALITY OF ESTIMATES

Table 2 contains the probability score, bias, slope, and scatter measures for all two-month and six-month judges. Figures 3a, b, and c depict these measures in covariance graph form.

In general, the physicians were superior to the other two groups in their overall PS. However, the physicians were not superior to the other groups on every component of the PS. One interesting example is bias. Consider first the two-month estimates. Reference to figure 3a reveals that the patients' average two-month estimate ($\bar{f}$) and the two-month base-rate of survival ($\bar{d}$) crossed almost exactly on the 45-degree line. This means that the patients' estimates had negligible bias, which is confirmed by the data presented in table 2. Figure 3b shows that surrogates were somewhat pessimistic. This corresponds to the bias of -0.037 listed in table

*The 692 included patients did differ significantly from certain subsets of the excluded patients. The 692 included patients had a significantly ($p < 0.05$) higher six-month survival rate (70.2%) than the 402 patients who were interviewed but did not give a two-month estimate (59.0%). Included patients also had a significantly higher treatment intensity score (TISS)¹⁹ (17.1) than the 83 patients whose physicians and surrogates both failed to make prognostic estimates (14.6), and they were significantly younger (60.8 years) than two groups: the 402 patients who were interviewed but did not give a two-month estimate (65.0 years) and the 148 patients whose physicians made no six-month estimate (62.2 years). Also, it should be noted that this report was written before the completion of the SUPPORT study. Therefore the small proportion of patients whose final outcomes were unknown at the time this report was written will eventually be included in one of the outcome categories.

Table 1 • Disease Categories of Patients Who Gave Two-month Estimates and Patients Who Gave Six-month Estimates

Disease Category	2-month Estimators	6-month Estimators
Acute respiratory failure	116	106
Chronic obstructive pulmonary disease	83	76
Congestive heart failure	199	183
Cirrhosis	79	74
Coma	1	1
Colon cancer	66	54
Lung cancer	86	69
Multiple-organ-system failure + malignancy	23	19
Multiple-organ-system failure + sepsis	39	34
TOTAL	692	616

2. Figure 3c shows that the physicians were more pessimistic, as evidenced by the fact that the $\bar{f}$ and $\bar{d}$ lines cross substantially beneath the 45-degree line. In fact, the physicians were approximately 12% too pessimistic in their two-month estimates.

Notice that whereas the physicians were too pessimistic when their estimates are viewed in the aggregate, their slope indicates that they were far more discriminating than either the patients or the surrogates. This is easily seen by comparing the slopes of the lines connecting $\bar{f}_1$ and $\bar{f}_0$ in figures 3a, b, and c. The patients have a much flatter slope than either the surrogates or the physicians, which is again confirmed in the data presented in table 2.

Another index of estimation quality on which the physicians were not clearly superior to the other two groups was "scatter," which is a measure of estimation variability unrelated to patients' actual survival. A comparison of the patterns of the histograms in figures 3a, b, and c suggests that the surrogates had more scatter in their estimates than did the other two groups, which in turn were similar to each other. Consulting Table 2 confirms that the physicians' two-month estimates were slightly more variable than those of the patients

Table 2 • Mean Probability Score ($\overline{PS}$), Mean Estimate ($\bar{f}$), Mean Estimate Assigned to Survivors ($\bar{f}_1$), Mean Estimate Assigned to Non-Survivors ($\bar{f}_0$), Bias, Slope, and Scatter for All Two-month and Six-month Estimators*

	$\overline{PS}$	$\bar{f}$	$\bar{f}_0$	$\bar{f}_1$	Bias	Slope	Scatter
Two-month estimates ($n = 692$)							
Patients	0.146	0.864	0.790	0.877	0.017	0.087	0.038
Surrogates	0.138	0.810	0.648	0.839	-0.037	0.191	0.052
Physicians	0.134	0.726	0.549	0.758	-0.121	0.209	0.039
Six-month estimates ($n = 616$)							
Patients	0.234	0.832	0.745	0.870	0.134	0.125	0.054
Surrogates	0.206	0.758	0.604	0.824	0.060	0.220	0.075
Physicians	0.180	0.592	0.412	0.670	-0.106	0.258	0.053

*Base rate $\bar{d} = 0.847$ for two-month survival. Base rate $\bar{d} = 0.698$ for six-month survival.

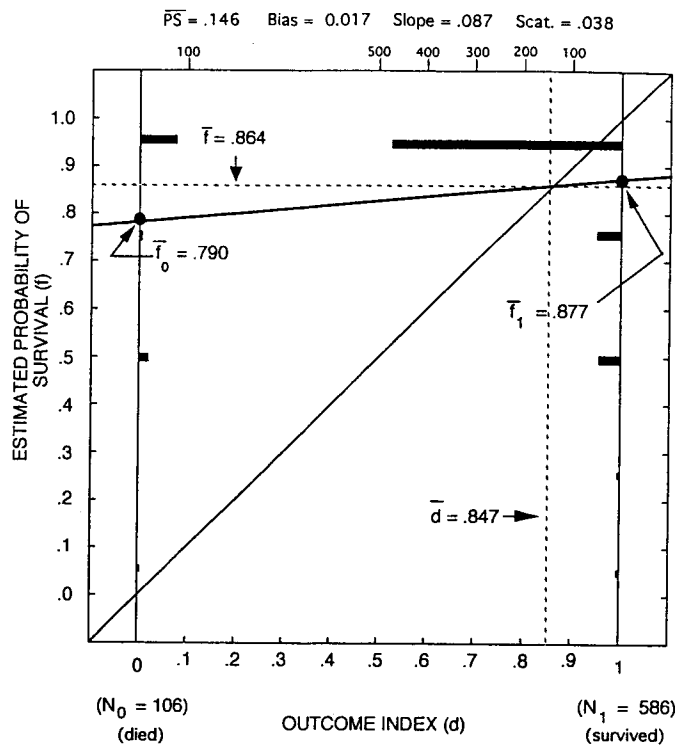


FIGURE 3a. Covariance graph for patients who made two-month estimates.

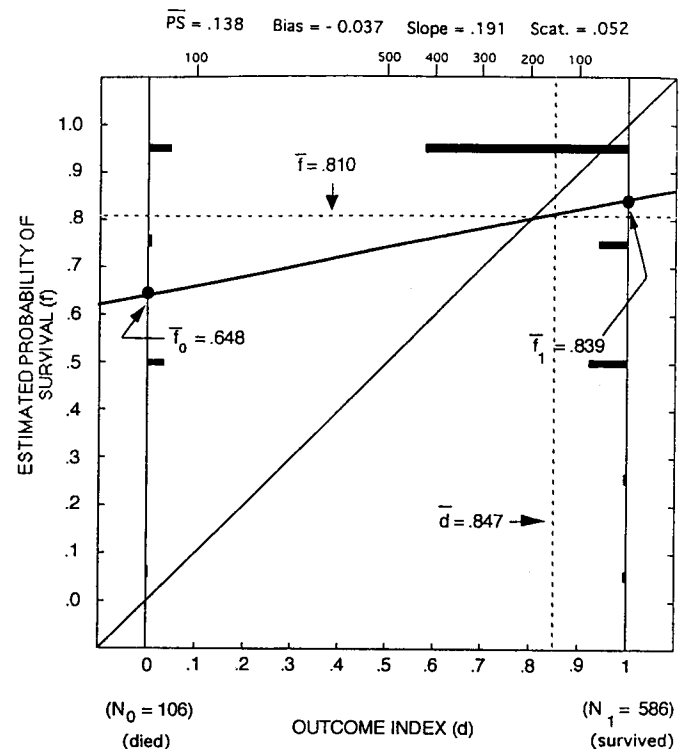


FIGURE 3b. Covariance graph for surrogates who made two-month estimates.

but less variable than those of the surrogates.

The histograms along the left and right margins of the panels in figure 3 depict the actual numbers of estimates made in the individual categories. These graphs indicate that the patients and the surrogates placed a large majority of their estimates in the most optimistic category. Such behavior is particularly detrimental to slope. The patients' data show inferior slope, because the patients who survived and those who did not *both* tended to place the bulk of their estimates in the highest category ($\geq 90\%$). Thus, the mean estimate of survival made by those who did survive exceeded the estimate of those who did not survive by only 0.087 ($0.877 - 0.790$).

The histograms in figure 3c, which depict the physicians' data, show that they used the less optimistic categories for the non-surviving patients more than those patients themselves did (fig. 3a), resulting in greater slope.

The histograms also provide valuable information with regard to the source of the physicians' negative bias. Since the physicians used the less optimistic categories more than did the other two groups and were frequently incorrect for doing so, the physicians' bias was much more negative than the biases of the other two groups.

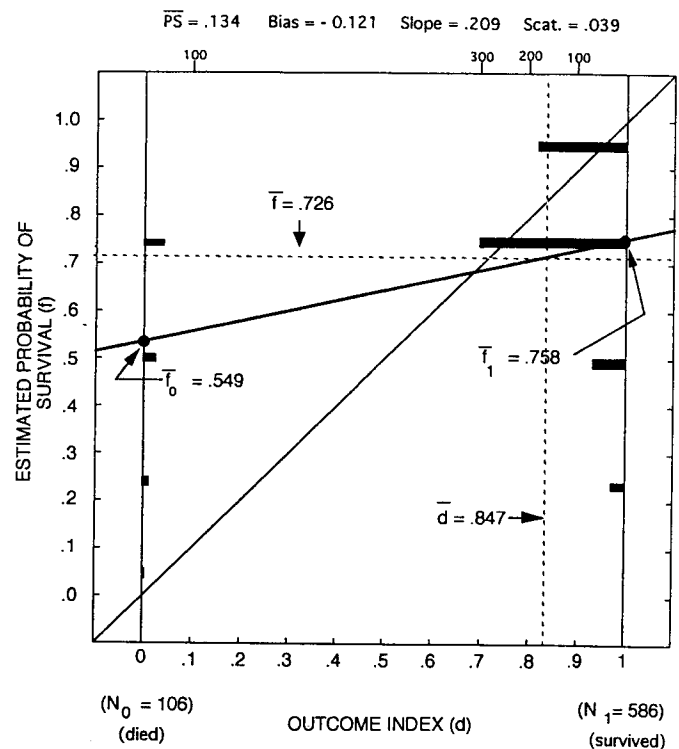


FIGURE 3c. Covariance graph for physicians who made two-month estimates.